

Name		Date of Performance	
Roll No.		Date of Submission	
Branch		Examined	

Experiment No.

TITLE:- STUDY AND APPLICATIONS OF PROFILE PROJECTOR AND TOOL MAKER'S MICROSCOPE

AIM:- To study profile projector and Tool Maker's Microscope

OBJECTIVES:-

1. To study Profile Projector
2. To construct magnified drawing of object on tracing paper
3. To measure component
4. To measure deviation from actual size
5. To study Tool Maker's Microscope

Apparatus:-

INSTRUMENTS
Profile Projector
Tool Maker's Microscope

SPECIFICATIONS
10x, 20x, 50x
10x

THEORY:-

PROFILE PROJECTOR:-

By using lenses and beams of light, profiles of small shapes can be magnified. The enlarged image can be compared with accurate drawings made to the scale of magnification. Such a comparison can reveal any deviation in the size and contours of the objects and to get a numerical assessment of such deviation. Such measurement can be made on the enlarged shadow. The measured dimensions on the shadow will then have to be divided by multiplication factor. The measured dimension or the shadow will then have to be divided by multiplication factor. The projection apparatus used for this purpose is termed as an Optical Profile Projector.

Fig. shows the optical arrangement in a Profile Projector 'A' is a point source of light kept at the focus of the collimating lens 'B'. the collimating lens

sends the light in a parallel direction and then sends through the focus 'E' by the projection lens 'D' inverted real image is formed and can be seen on the screen placed at 'F'.

TOOL MAKER'S MICROSCOPE:-

It is a precision inspection instrument. Fig. shows a block diagram of Tool Maker's Microscope. A heavy hollow base 'A' houses the lighting unit. Above the base is mounted the table 'B' which is supported on balls 'C' to provide the movement about vertical axis. The blade can also be moved in horizontal plane in two direction perpendicular to each direction as shown by 'e' and 'f'. this movements are controlled by two micrometer drums projecting up from the back side of the base is column 'b' which can be rotated about axis 'XX'. The column carries the microscope unit 'h' which can be adjusted up and down. The column on guide ways 'j' can be clamped by knob 'k' at the top of the microscope. The eyepiece unit I carries different eyepieces. Any one of which can be used at time of projection. Attachment can be fitted to the unit to obtained a shadow, on a screen light from the lamp in the back of the base taken through a system of lenses and reflectors, goes up through the central transparent part of table into the microscope.

Work is supported between centres, or on V blocks or placed simply on flat glass table micrometers having range of 25mm and read to 1micron.

PROCEDURE:-

1. Set the jaw in the centre of field and square to optical axis.
2. Select a magnification that will give a clear image.
3. Tabulate the dimension.
4. Draw projection drawing on tracing paper using usual drawing instruments.
5. Make upper and lower of tolerance dimension in the drawing.
6. Match the following drawing with image.
7. Measure the error.

OBSERVATION TABLE:-

Sr. No	Features	Controlled Dimensions	Actual Dimensions	Error Observed
Pressure Angle				
Thickness of tooth				

CONCLUSION:-